

Quiz3 - Results



Attempt 1 of 1

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Attempt Score 9 / 10 - 90 %

Overall Grade (Highest Attempt) 9 / 10 - 90 %

Question 1

1 / 1 point

In an LSTM, what do the gates take as input?

- ☐ x_{t-1} and h_t
- ✓ ☒ x_t and h_{t-1}
- ☐ x_t and h_t
- ☐ x_{t-1} and h_{t-2}

Question 2

1 / 1 point

When do RNNs face the vanishing gradient problem?

- ☐ During forward propagation
- ☐ During loss calculation
- ✓ ☒ During back propagation



All of the above

Question 3

1 / 1 point

What is the main difference between sequence labelling (Seq-L) and sequence generation (Seq-G)?



In Seq-L, the output can be of a different length than the input, but that's not true for Seq-G



There is no difference, both of them require the output to be of a different length than the input



In Seq-G, the output can be of a different length than the input, but that's not true for Seq-L



There is no difference, both of them require the output to be of the same length as the input

Question 4

1 / 1 point

Which of the following is a sequence labeling task?



Question Answering



POS Tagging



Machine translation



Document summarization

Question 5

1 / 1 point

What are the gates present in an LSTM?

- ☐ update gate and reset gate
- ☐ input gate, reset gate and output gate
- ☐ input gate and output gate
- ✓ ☒ input gate, output gate and forget gate

Question 6

0 / 1 point

Why is beam search helpful during decoding an output sequence?

- ☐ It allows us to find the most optimal output sequence
- ➡ ☐ It is more efficient than an exhaustive search
- ✗ ☒ Both of the above
- ☐ None of the above

Question 7

1 / 1 point

What is the purpose of adding the attention mechanism to the encoder decoder architecture?

- ☐ The encoder and decoder have different embedding sizes, and the attention mechanism acts as the necessary intermediate weight matrix
- ✓ ☒ The final encoder hidden state is a single vector which is incapable of effectively capturing all the information in the source sequence
- ☐ Lack of attention mechanism can cause exploding gradients



All of the above

Question 8

1 / 1 point

Which of the following functions are used in the mathematical implementation of the attention mechanism in an LSTM?



tanh



softmax



ReLU



sigmoid

Question 9

1 / 1 point

Which of the following tasks is true for a recurrent neural network?



It can handle any length of input, and it handles all sequence tokens in parallel



It can handle any length of input, and it handles the sequence tokens one by one



It can only handle a maximum sequence length equal to the batch size



It can handle any length of input, and it handles the sequence tokens in multiple fixed windows

Question 10

1 / 1 point

Which of the following can be added to improve upon an RNN?

- ☐ Character level embeddings
- ☐ Adding a bidirectional component
- ☐ Gates that control information flow
- ✓ ☒ All of the above

Done